creating, manipulating, and running quantum circuits. Google's Quantum Computing Service uses the Quantum Engine API for submitting programs to quantum processors, although hardware access is typically restricted to approved users or research partnerships.

Google's contribution is visible through Cirq, Colab notebooks, tutorials, and published research. Google's work has also influenced discussions on quantum supremacy, error correction, and scalable quantum architectures. While not as openly accessible to the public as some other platforms, it remains important in research-oriented quantum computing.

Amazon Braket represents a different model. Rather than focusing on one hardware technology, it provides a managed service for accessing multiple quantum hardware providers and simulators through a unified AWS experience. Users can work with superconducting, trapped-ion, and neutral-atom systems, depending on current provider availability.

Braket also integrates with familiar AWS services such as identity management, storage, monitoring, and billing. This makes it valuable for enterprises that already use AWS and want to experiment with quantum computing without creating a separate operational model.

Microsoft Azure Quantum also takes an aggregator and ecosystem-oriented approach. It provides access to quantum hardware and simulators from multiple providers and supports development through tools such as Q#, Python integrations, and Azure-based workflows. Azure Quantum is often positioned within a broader strategy that combines high-performance computing, artificial intelligence, materials science, chemistry, and quantum readiness.

Beyond these large platforms, several other players contribute to access and diversity. D-Wave offers cloud access to quantum annealing systems and hybrid solvers, which are particularly relevant for optimization-style problems. IonQ, Quantinuum, Rigetti, Pasqal, QuEra, IQM, and other hardware companies provide access either directly or through cloud marketplaces.

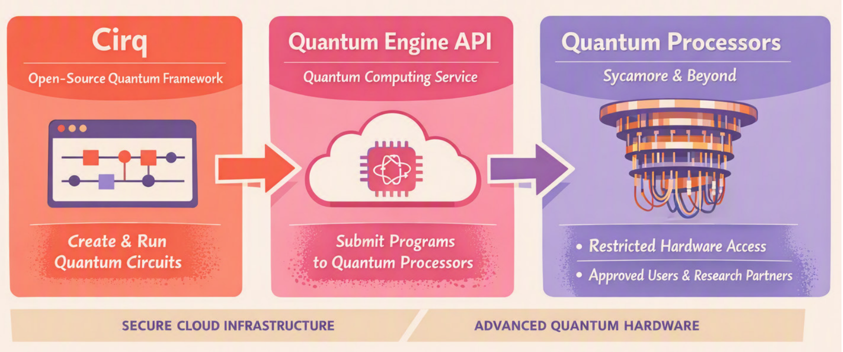

Figure 2: Google's quantum ecosystem

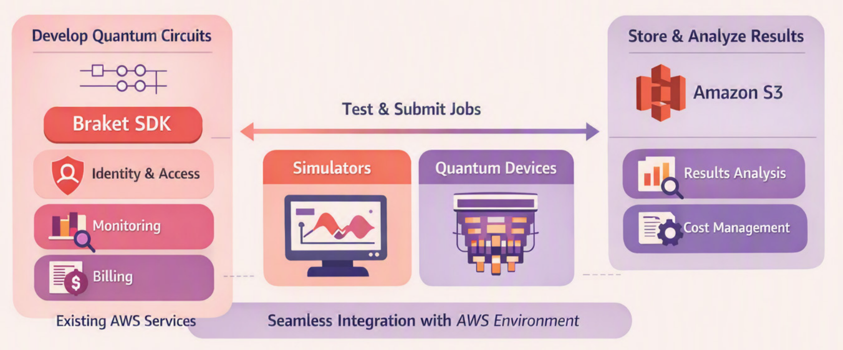

Figure 3: AWS Braket integration with AWS native services

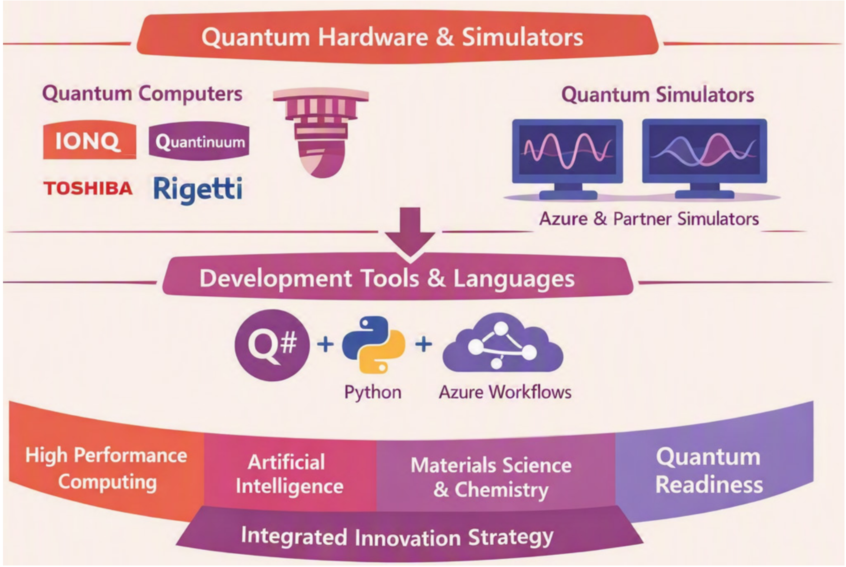

Figure 4: Microsoft Azure's aggregated quantum ecosystem

Getting started: User access and interface

Getting started with cloud quantum computing is far easier today than it was a decade ago. A user typically begins by creating an account on